

# MACHINE LEARNING IN PHYSICS FOUNDATIONS 1

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PHY6937 / PHY4936

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# Goals of this Course

1. Gain a good understanding of the mathematical basis of machine learning (ML), which is a foundational component of artificial intelligence (AI).
  2. Gain experience building ML models using **PyTorch** to solve data science problems in physics.
  3. Gain experience with different ML models.
  4. Gain an understanding of AI agents.
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# Course Evaluation

In addition to your **Course Participation** (attending, asking questions, contributing to discussions), your course grade will be based on:

1. **In-class Assignments**
  2. **Project with a write-up and a brief presentation**
    1. Graduates: A project related to your research area.
    2. Undergraduates: A project selected from a few options.
    3. Presentations: Last week of the semester.
  3. **Grade**: Graduates (S/U), Undergraduates (Letter Grade).
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# What is Artificial Intelligence?

## **Artificial Intelligence**

Algorithms that cause machines to exhibit *human-* or *superhuman-*level intelligence.

## ChatGPT



**Developer(s)** OpenAI

**Initial release** November 30, 2022

**Stable release** August 7, 2025

**Engine** GPT-5

Large Language Model (LLM)

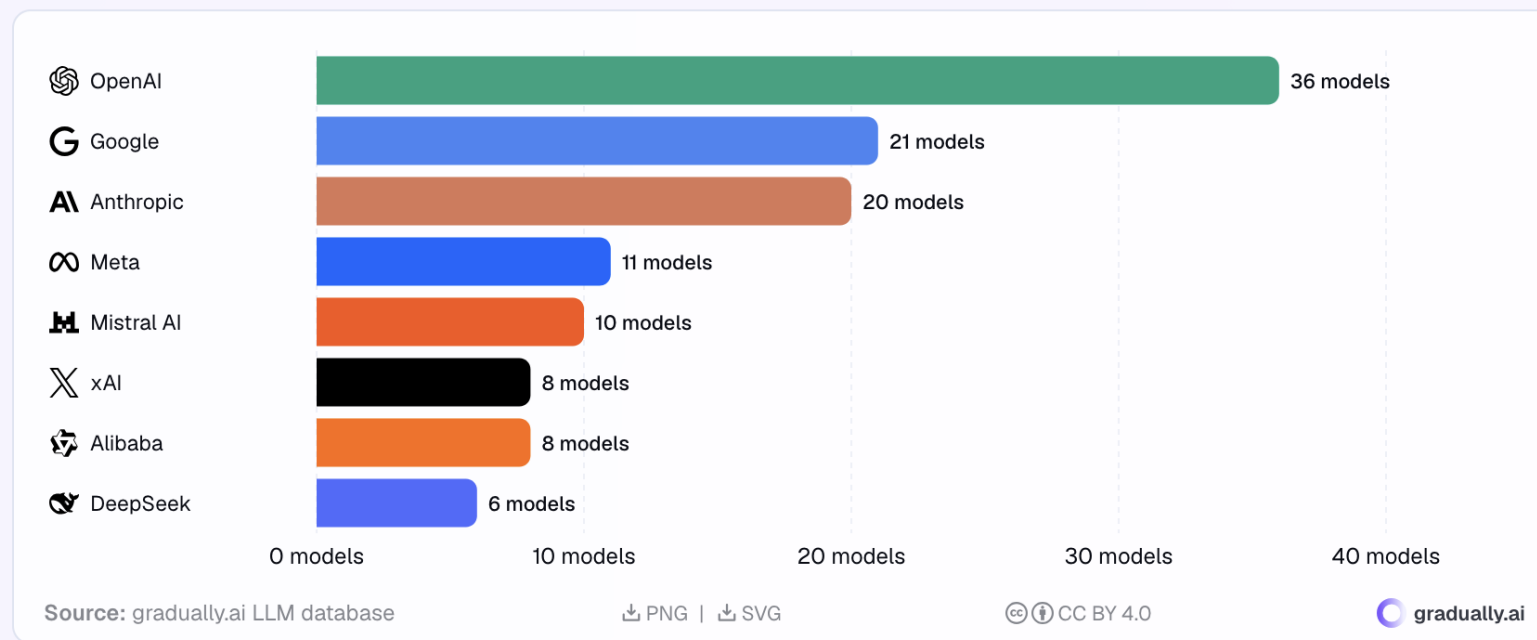


BY ANTHROPIC

Large Language Model (LLM)

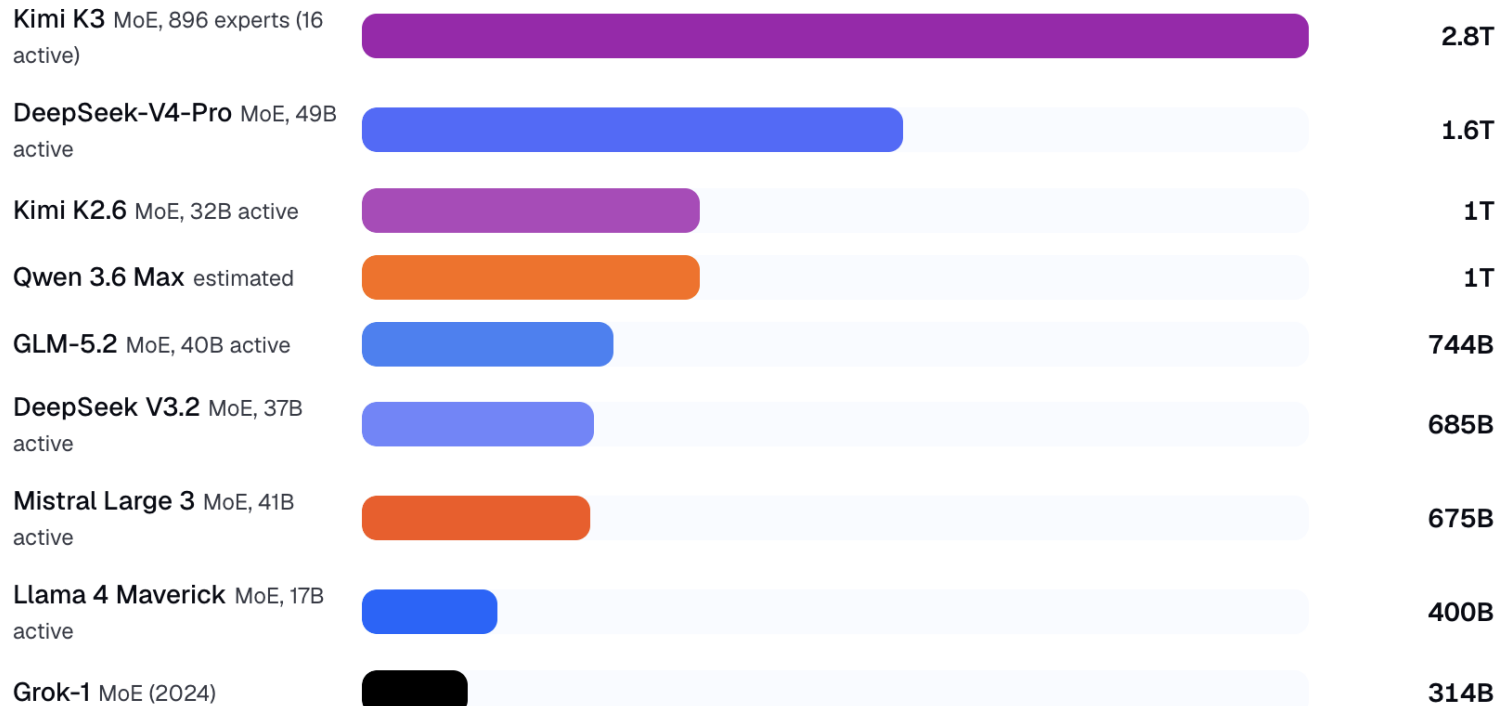
## 2. The Biggest LLM Providers by Model Count

A simple indicator of how active a lab is: the number of models it maintains. The chart below shows how many of the models we track belong to each provider:



<https://www.gradually.ai/en/llm-statistics/>

# Model sizes: Number of parameters



Source: gradually.ai LLM database

📄 PNG | 📄 SVG

CC BY 4.0

 gradually.ai

<https://www.gradually.ai/en/llm-statistics/>



# Claude Demo

# What is Machine Learning?

## **Artificial Intelligence**

Algorithms that cause machines to exhibit human- or *super-human* level intelligence.

## **Machine Learning**

Algorithms for modeling data.

# What is Machine Learning?

## Supervised Learning

**Data:**  $(x, y)$

$y$  are labels

**Task:**  $x \rightarrow y$

**Use cases:**

- Classification, regression, translation, etc.

## Unsupervised Learning

**Data:**  $x$

no labels

**Task:** find structure in,  
and/or model, data

**Use cases:**

- Clustering, data compression, solving differential equations, etc.

# What is Machine Learning?

## Generative Learning

**Data:**  $x$

may or may not be associated with labels

**Task:**  $x \rightarrow p(x) \rightarrow x$

**Use cases:**

- fast simulators, image/text generation, chatbots, etc.

## Reinforcement Learning

**Data:**  $(x, a, r)$

$x$  state of the environment

$a$  action taken on environment

$r$  reward arising from action

**Task:** find optimal  $x \rightarrow a$

**Use cases:**

- Game playing, robotics, accelerator controls, etc.

# What is Deep Learning?

## **Artificial Intelligence**

Algorithms that cause machines to exhibit human- or *super-human*-level intelligence.

## **Machine Learning**

Algorithms for modeling data

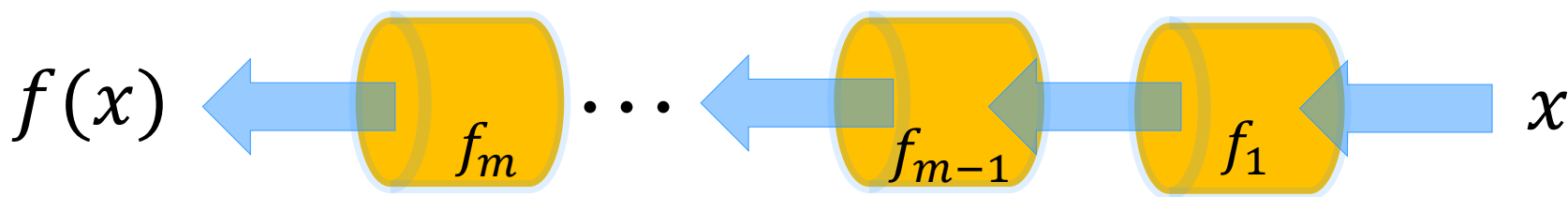
## **Deep Learning**

ML using (large) neural networks

# What is Deep Learning?

Deep learning is the science and art of fitting models to data using functions formed through *composition*:

$$\begin{aligned} f(x) &= f_m \circ f_{m-1} \circ \cdots f_1 \\ &= f_m(f_{m-1}(\cdots f_1(x) \cdots)) \end{aligned}$$



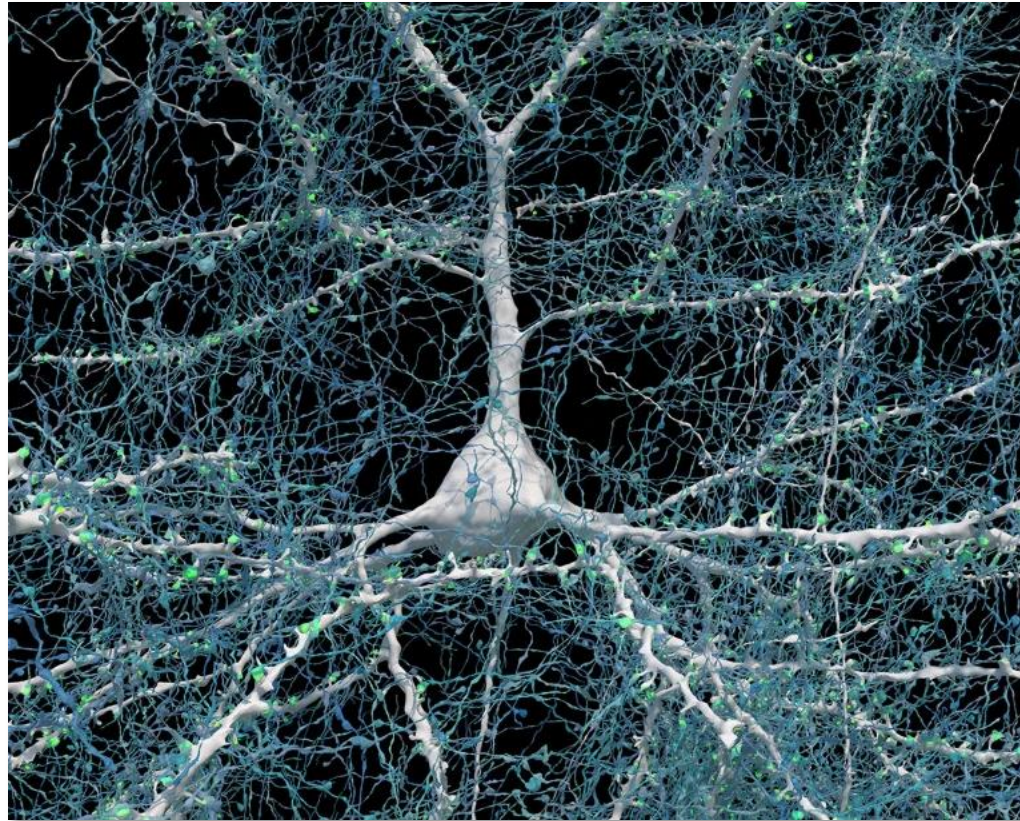
Each of these functions is referred to as a **layer**. Kimi K3, from Moonshot AI, has **93** layers and **2.8 trillion** parameters!

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# **BASIC BUILDING BLOCK: THE PERCEPTRON**

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# The Brain's Computational Unit



<https://www.nature.com/articles/d41586-024-01387-9>



# Basic AI Comp. Unit: The Perceptron

$$y = g(xA^T + b)$$

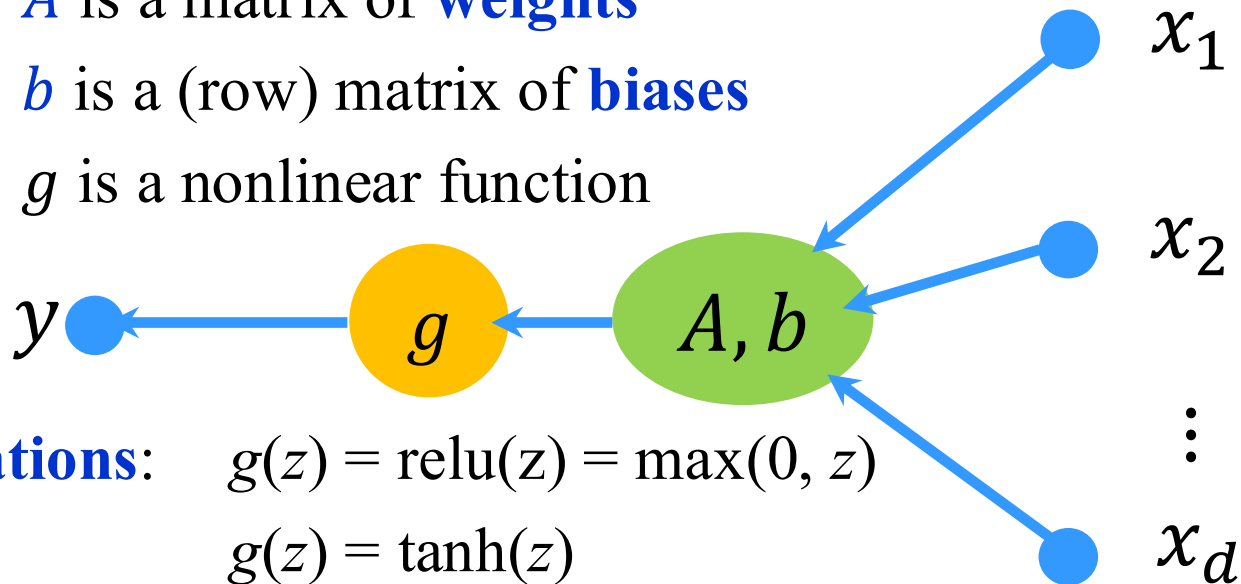
(Frank Rosenblatt, 1958)

$x$  is a (row) matrix of input data

$A$  is a matrix of **weights**

$b$  is a (row) matrix of **biases**

$g$  is a nonlinear function



**Activations:**  $g(z) = \text{relu}(z) = \max(0, z)$

$g(z) = \tanh(z)$

$g(z) = \text{sigmoid}(z) = 1 / (1 + \exp(-z))$

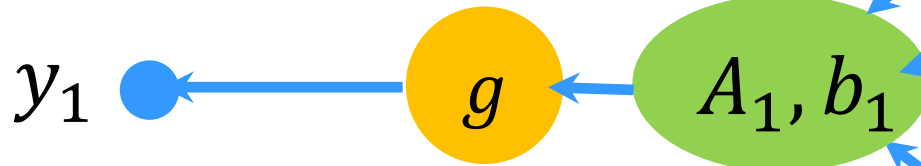
<https://news.cornell.edu/stories/2019/09/professors-perceptron-paved-way-ai-60-years-too-soon>

# MULTI-NODE PERCEPTRON

# Multi-Node Perceptron

$$g(z) = \text{relu}(z)$$

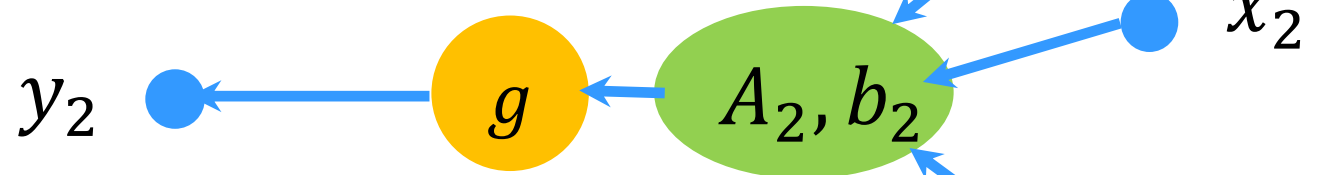
$$y_1 = g(xA_1^T + b_1)$$



**Example:**

$$A_1 = (2, -3, 1), \quad b_1 = -5$$

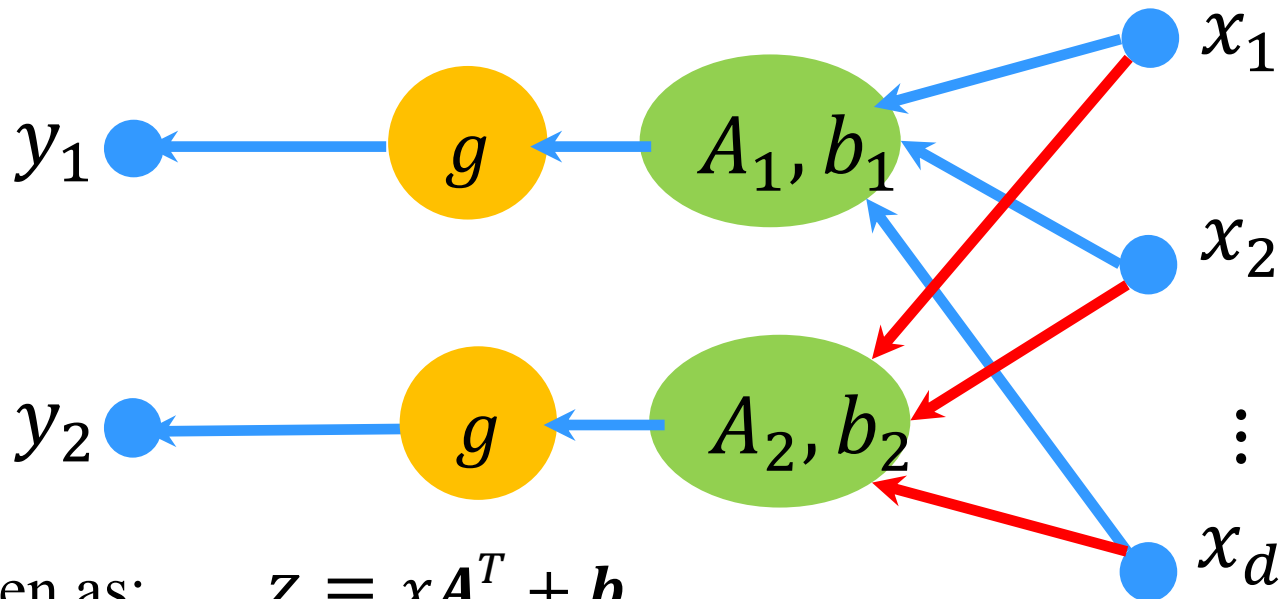
$$y_2 = g(xA_2^T + b_2)$$



$$A_2 = (1, 2, 3), \quad b_2 = -4$$

# Multi-Node Perceptron

A multi-node perceptron is often drawn as follows,

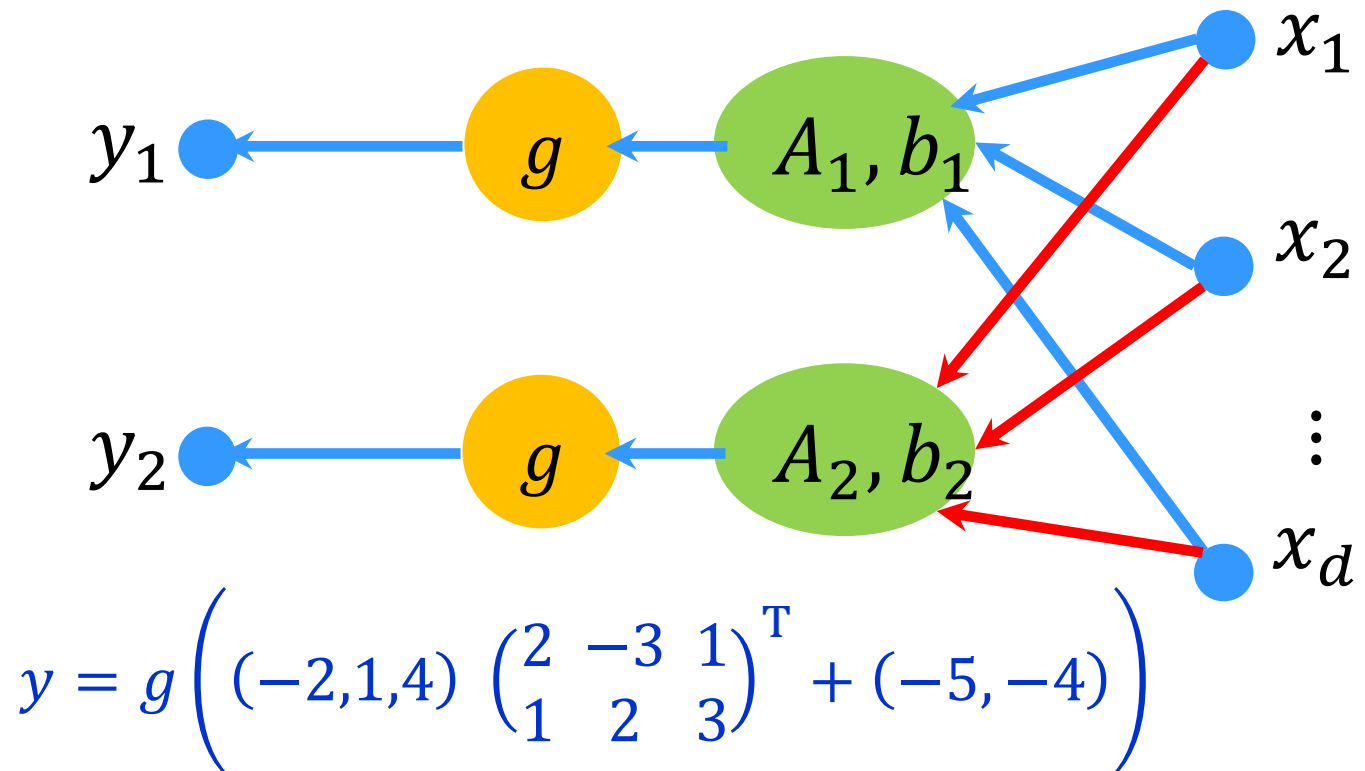


Written as:  $z = xA^T + b$

with  $y = g(z)$ , usually, the applied *elementwise*,  
that is, to every element of its matrix input.

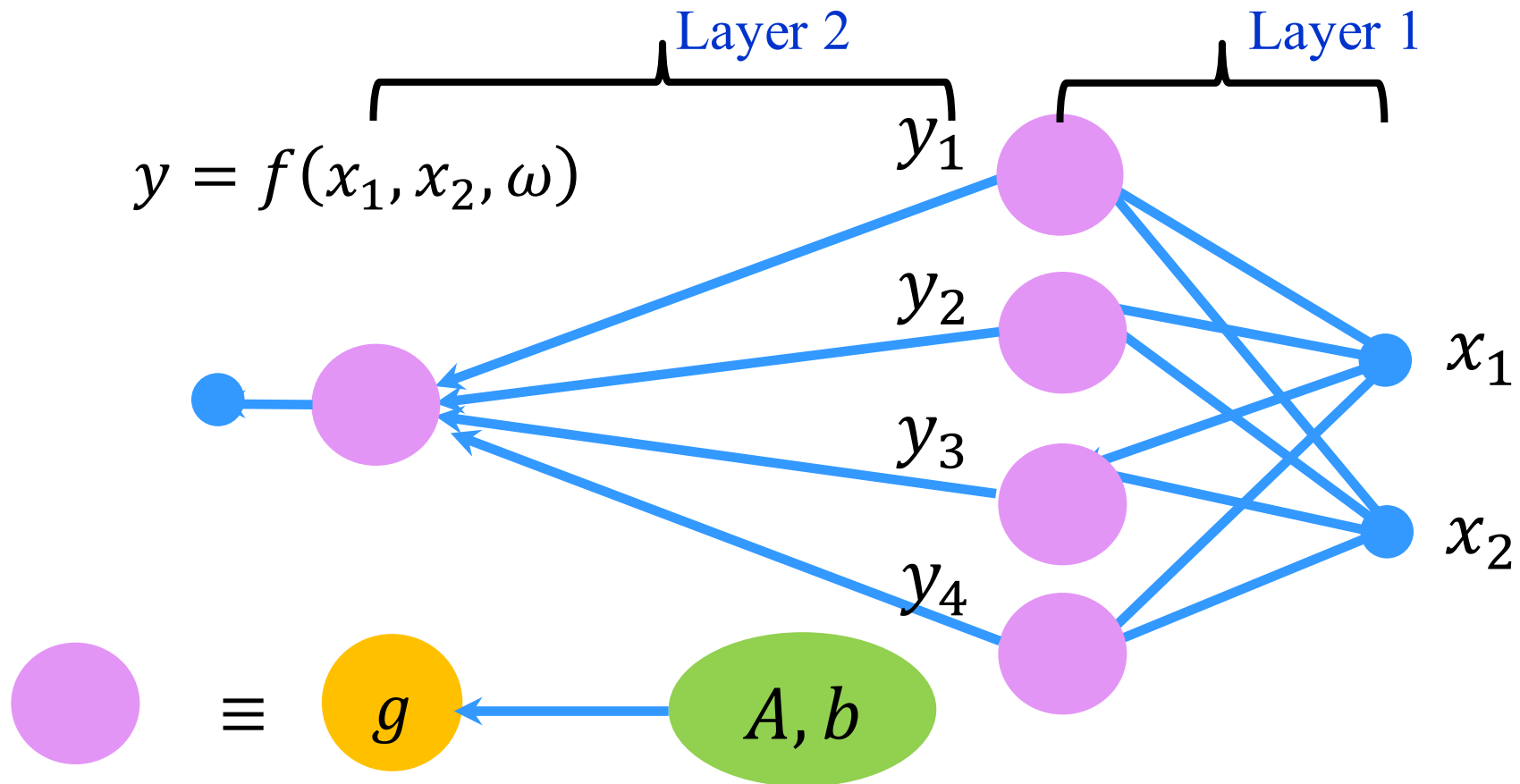
# Multi-Node Perceptron: Example

$$x = (-2, 1, 4), \quad A = \begin{pmatrix} 2 & -3 & 1 \\ 1 & 2 & 3 \end{pmatrix}, \quad b = (-5, -4)$$



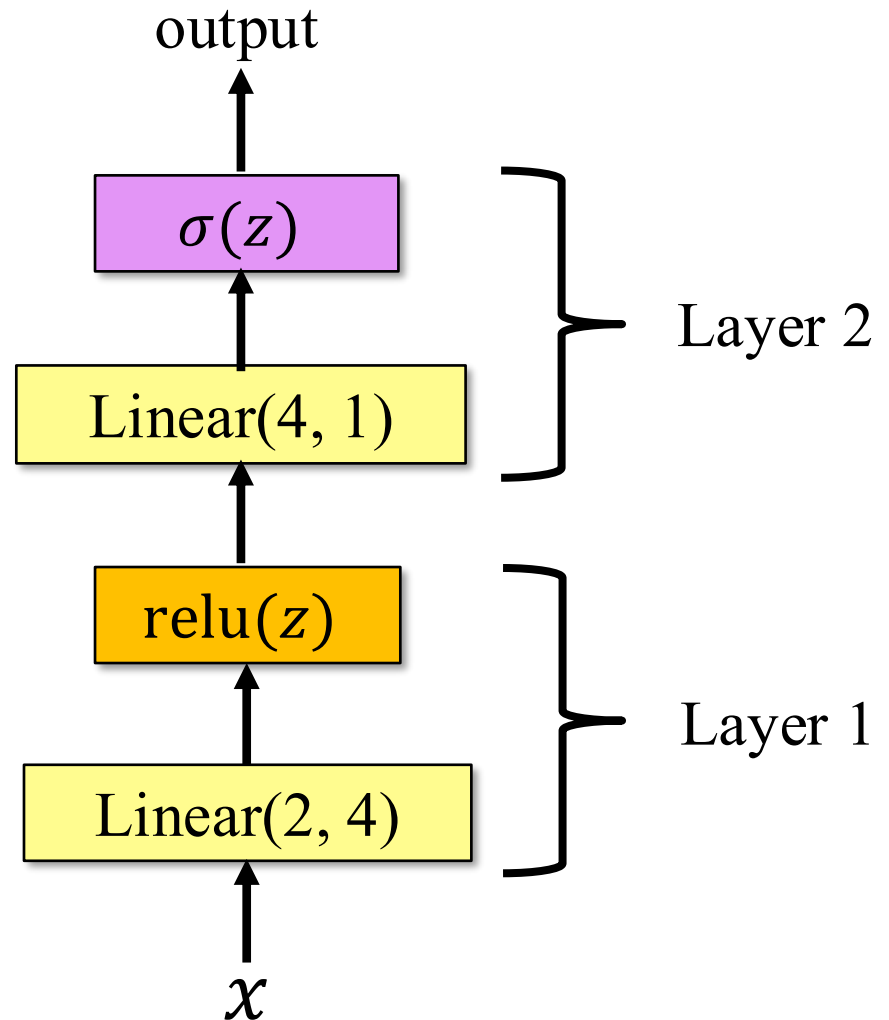
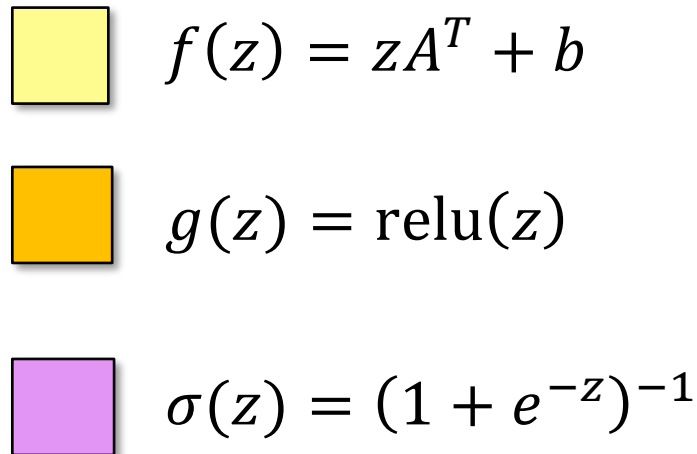
# Multi-Node Perceptron

Consider modeling the data triplets  $D = \{(x_1, x_2, y)_{i=1}^N\}$  with a function of the form  $y = f(x_1, x_2, \omega)$ .



# Multi-Node Perceptron

Given the complexity of ML models (i.e., ML functions), it is now common to use a graphical representation that uses higher-level components. For example,

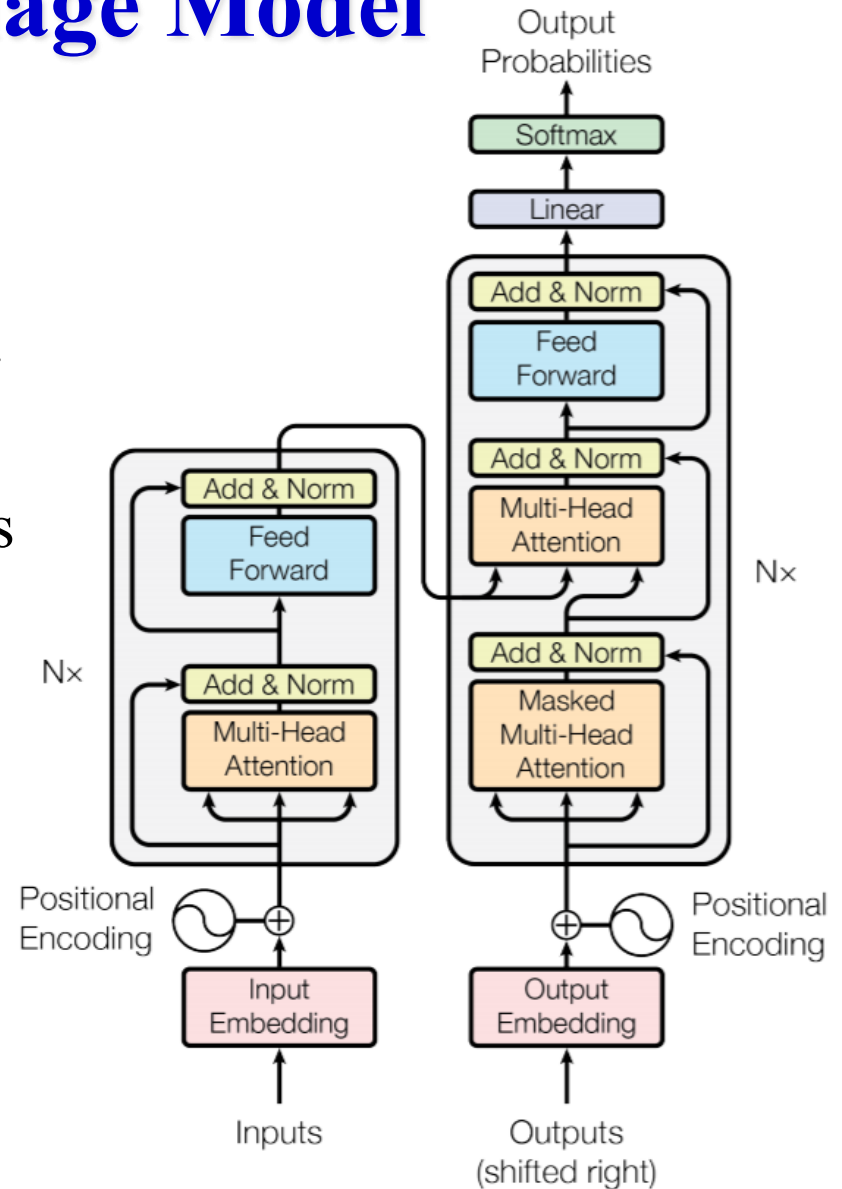


# Large Language Model

Here, for example, is high-level graphical representation of an ML model called a **transformer**.

In ChatGPT 3, there are 96 layers with the same architecture, an instance of which is depicted as the two vertical grey boxes.

(Notice, the color-coding differs from ours!)





# Summary

## ➤ General Approaches

- Supervised, unsupervised, generative, and reinforcement learning.

## ➤ Deep Learning

- Uses functions constructed through deep composition.

## ➤ The Perceptron

- Basic computational unit: matrix multiplication and addition followed by a nonlinear map.
- We described the multi-node perceptron and how machine learning models are represented graphically.